Reviewer #1

Questions

1). The paper addresses a culturally significant and underexplored problem by focusing on keyframe extraction for Kathakali dance videos. The proposed STAKE framework effectively combines CNN-based spatial feature extraction, CLIP-based semantic understanding, and Bi-LSTM-based temporal modelling, and the paper is generally well organized with a clear description of the methodology and dataset collection process.

Justification: We thank the reviewer for their positive assessment of our work.

2). While the proposed architecture is interesting, the experimental evaluation is relatively limited. The dataset appears small (approximately 210 videos), and the paper does not compare STAKE against existing keyframe extraction methods on the same dataset. **Including baseline comparisons, statistical significance analysis, and cross-validation experiments would help demonstrate the effectiveness of the proposed approach more convincingly.**

Justification: In the revised manuscript, we have included baseline comparisons with keyframe extraction methods from the literature, including clustering-based and deep learning-based approaches. Furthermore, a statistical significance analysis has also been performed using Wilcoxon's signed rank test. The results of both the baseline comparisons and statistical significance analysis have been added to the Results, specifically section 4.3 (page number 15, Table 5) and section 4.5 (page number 16) respectively. The comparative results demonstrate that the proposed STAKE framework consistently outperforms the existing methods.

Regarding cross-validation, we initially explored frame-level cross-validation. However, since multiple frames originate from the same video, this resulted in data leakage, with frames from the same video appearing in both the training and testing folds, leading to overfitting. Therefore, we adopted a strict video-level train/validation/test split, ensuring that all frames from a given video belong exclusively to one subset. This information has been updated in the Experimental setup section, specifically section 4.1 (page 12)

Additionally, we acknowledge that our dataset is small with only 210 videos. We are in the process of building a bigger dataset with more words as well as with more videos, taken from different people, to expand our evaluation. We are planning to publish this expanded work as well as the bigger dataset in the future. This has been added to the manuscript in the Conclusion and Future Work section, specifically the last paragraph in section 5 (page 17)

3). A key concern is the lack of ablation studies. The paper claims that combining CNNs, CLIP, and Bi-LSTM improves performance, but there is no experimental evidence showing the contribution of each component individually. **Evaluating variants such as CNN-only, CLIP-only, CNN+Bi-LSTM, and CLIP+Bi-LSTM would provide stronger support for the design choices and help quantify the importance of each module.**

Justification: An ablation study has been performed and added to the revised manuscript in the Results section, specifically section 4.4 (page number 15, Table 6). The study evaluates four variants of the proposed framework, namely CNN-only, CLIP-only, CNN+Bi-LSTM, and

CLIP+Bi-LSTM, to quantify the individual contribution of each component. The experimental results show that though each component has a good performance, the complete STAKE model achieves the highest performance among all evaluated variants.

Reviewer #2

Questions

1. **The novelty and key technical contribution of the proposed work are not sufficiently clear.** The framework mainly combines existing components, including CNN backbones, CLIP-based feature extraction, attention mechanisms, and Bi-LSTM networks.

Justification: We regret to hear that the novelty and technical contribution were not clear. The main contribution of this work is in addressing the automatic keyframe extraction problem for Kathakali dance videos, where the objective is to distinguish keyframes from transition frames. The proposed work is designed as the first stage of our larger pipeline for translating Kathakali dance videos into their corresponding semantic meanings. By automatically identifying keyframes while eliminating transition frames, our method can significantly reduce noisy frames and enable efficient video classification of Kathakali dance postures. This has been added to the Introduction section, specifically last paragraph in Section 1 (page number 3), in a more detailed manner.

2. While the application of Kathakali video analysis is interesting, the manuscript does not clearly demonstrate a methodological advancement beyond the integration of established techniques. The authors should also follow standard benchmarking for the same.

Justification: We have included an ablation study with variants such as CNN-only, CLIP-only, CNN+Bi-LSTM, and CLIP+Bi-LSTM, to identify the contribution of each component in the proposed architecture. We have also performed comparisons with several benchmark keyframe extraction approaches reported in literature review, such as, Nearest Neighbour Clustering, K-Medoids Clustering, Cosine similarity with SSIM, SVM and CNN-based methods, etc. These additional experiments have been incorporated into the revised manuscript under the Results section, with the benchmark comparisons included in subsection 4.3 (page 15, Table 5) and Ablation Studies in Section 4.4 (page 15, Table 6). The comparative results demonstrate that the proposed STAKE framework consistently outperforms the existing methods.

Reviewer #3

Questions

1. The paper proposes a keyframe extraction framework for Kathakali dance videos using CNN features, CLIP-based semantic features, and Bi-LSTM temporal modelling. The application is interesting, and the dataset creation effort is valuable, as dance videos with expressive gestures

present a challenging keyframe extraction problem. Extracting meaningful frames from such videos can support applications like digital preservation, dance learning, video summarization, and retrieval of important gestures or expressions. **However, the paper could better explain how the proposed system would be useful in these practical scenarios.**

Justification: We would like to thank the reviewer for their remark on the application. We have revised the manuscript to better explain the practical applications of the proposed keyframe extraction framework. The proposed system is designed as the first stage of our larger pipeline for translating Kathakali dance videos into their corresponding semantic meanings. These extracted keyframes can facilitate the digital preservation and archival of Kathakali performances by identifying key dance postures. They can also support dance education by helping students focus on the more significant postures, *Mudras*, and *Navarasas* during training, as well as be used for dance video generation and dance video interpretation. Furthermore, the framework can be applied to video summarization, efficient indexing and retrieval of dance sequences, and the development of intelligent tools for learning and analysis of classical dance. These practical applications have now been discussed in greater detail in the revised manuscript in the Introduction section, specifically last paragraph in Section 1 (page number 3).

2. The novelty mainly comes from the application domain rather than the individual model components. CNN, CLIP features, and LSTM-based temporal modelling are established techniques, **so the paper should better explain why this specific combination is effective for Kathakali keyframe extraction**. **The use of CLIP should also be described more accurately**, as CLIP provides semantic image-text representations rather than object detection. Additionally**, an ablation study is needed to understand the contribution of each component**, such as CNN features, CLIP features, and temporal modelling.

Justification: Kathakali keyframes consist of both generic visual information, such as body shape, postures, and spatial arrangement, as well as specific semantic information, such as Mudras (hand gestures) and Navarasas (facial expressions). Therefore, relying on a single model may not adequately capture the complexity of the dance postures. These have now been discussed in greater detail in the revised manuscript in the STAKE Model Architecture section, specifically Section 3.3 (page number 8).

We agree with the reviewer that CLIP is mainly used for sematic image-text representations rather than object detection, and have revised the manuscript accordingly. We are yet to explore which features are identified by CNN models and which ones by CLIP, which would be performed in future. This has been updated to the Conclusion and Future Work section, specifically the last paragraph in section 5 (page 17)

Furthermore, as suggested by the reviewer, we have included an ablation study with variants such as CNN-only, CLIP-only, CNN+Bi-LSTM, and CLIP+Bi-LSTM, to identify the contribution of each component in the proposed architecture. The results have been added under the Results section, specifically, section 4.4 (page 16, Table 6) which demonstrate that though each component contributes to the overall performance, the complete framework achieves the best results.

3. The experimental evaluation has some important limitations. Although the paper reports accuracy results, the precision, recall, and F1 scores are relatively low due to class imbalance between keyframes and transition frames. Since this issue is identified by the authors**, it would be better to address it using techniques such as weighted loss, sampling strategies, or focal loss rather than leaving it entirely as future work.** The evaluation strategy also needs more discussion, as selecting top-confidence keyframes and comparing them with ground truth keyframes may introduce bias. **More standard temporal evaluation metrics would provide a stronger validation.**

Justification: To investigate the effect of class imbalance, we experimented with both oversampling and undersampling strategies. However, oversampling increased the number of similar keyframe samples within the training set, thereby resulting in severe overfitting. Undersampling substantially reduced the number of training frames, leading to underfitting and degraded model performance. Based on these observations, we retained the original data distribution and additionally reported Weighted F1-score to provide a more balanced evaluation under class imbalance. This information has been updated in the Experimental setup, specifically the third paragraph in Section 4.1 (page 12)

Regarding the evaluation strategy, we agree that selecting only the highest-confidence keyframes may introduce bias. To address this concern, we have expanded the experimental evaluation by incorporating standard temporal evaluation metrics, including Mean Absolute Temporal Error (MATE) and Dynamic Time Warping (DTW). These metrics evaluate the temporal alignment between the predicted and ground-truth keyframes rather than relying solely on confidence-based selection, thereby providing a more comprehensive assessment of the proposed method. The results including MATE and DTW have been added to the Results, specifically section 4.2 (page 14, Table 4)

4. **The dataset is an interesting contribution, but additional details about the number of frames, keyframe distribution, and class imbalance statistics would help understand the difficulty of the task.**

Justification: We thank the reviewer for this valuable suggestion. To provide a clearer understanding of the dataset characteristics and the challenges associated with the keyframe extraction task, we have expanded the dataset description in the revised manuscript. Specifically, we have included a table summarizing the total number of frames and the corresponding number of manually annotated keyframes for the dataset under Manual Keyframes subsection. This additional information highlights the inherent class imbalance between transition frames and keyframes, thereby providing a better understanding of the complexity and difficulty of the proposed task. The details regarding the same have been added to the STAKE Model Architecture, specifically Section 3.3, pages 9 and 10, Tables 2 and 3)

Strengths

1) Addresses a unique and culturally relevant computer vision problem.
2) Dataset collection and manual annotation effort are valuable contributions.

3) The combination of spatial and temporal features is reasonable for video analysis.
4) The paper is well structured and easy to follow.

Justification: We thank the reviewer for the positive feedback given

Weaknesses

1) The motivation and practical use cases of keyframe extraction could be explained more clearly.

Justification: These practical applications have now been discussed in greater detail in the revised manuscript in the Introduction section, specifically last paragraph in Section 1 (page number 3).

2) The method is based on existing components and needs stronger novelty justification.

Justification: These have now been discussed in greater detail in the revised manuscript in the Introduction section, specifically last paragraph in Section 1 (page number 3).

3) CLIP usage is not described precisely (semantic features vs object detection).

Justification: We have corrected the usage of CLIP throughout the paper

4) Missing ablation study to validate the contribution of each module.

Justification: An ablation study has been added under the Results section, specifically, section 4.4 (page 16, Table 6)

5) Severe class imbalance affects performance and is not properly addressed.

Justification: This information has been updated in the Experimental setup, specifically the third paragraph in Section 4.1 (page 12)

6) The evaluation protocol may introduce bias due to top-confidence frame selection.

Justification: MATE and DTW have also been used as evaluation metrics and have been added to the Results, specifically section 4.2 (page 14, Table 4)

7) Dataset size is limited, so generalization claims should be made carefully.

Justification: We have removed the generalization claims and acknowledge that the dataset size is limited. This has been added to the Conclusion section, specifically, the last paragraph in section 5 (page 17)

8) Adding ablations, improving the evaluation protocol, addressing class imbalance, and clarifying the practical motivation would significantly strengthen the paper.

Justification: We would like to thank the reviewer for their feedback. As per the reviewer's suggestions, evaluation protocol has been improved using MATE and DTW evaluation in Results, specifically section 4.2 (page 14, Table 4). Information on the class imbalance has been added to the Experimental setup, specifically the third paragraph in Section 4.1 (page 12), and practical motivations have been discussed in greater detail in the revised manuscript in the Introduction section, specifically last paragraph in Section 1 (page number 3).